MATH 151 Lab 8

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```
import sympy as sp
from sympy.plotting import (plot,plot_parametric)
```

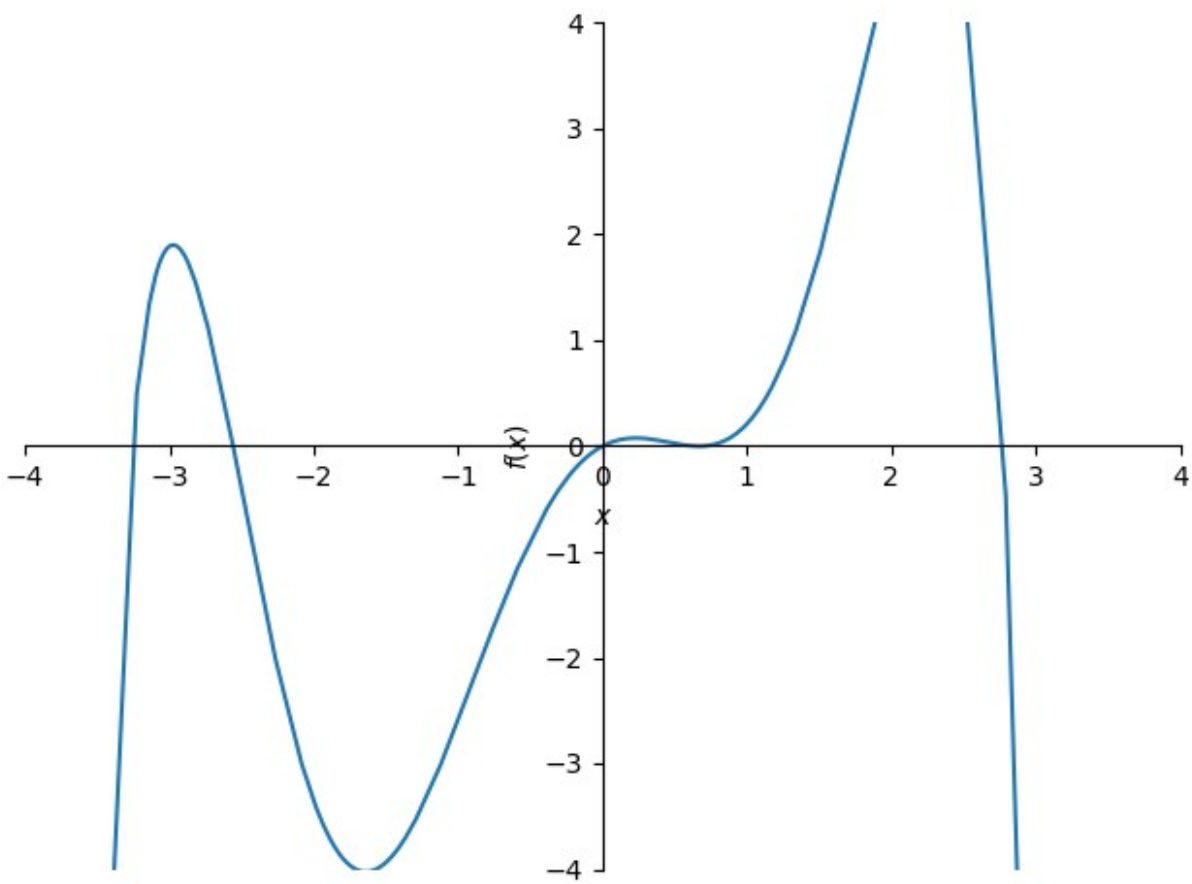
Question 1

1a

```
# Define vars and functions
x = sp.Symbol("x", real=True)
f = (-x**7 - 2*x**6 + 16*x**5 + 20*x**4 - 64*x**3 + 36*x**2 - 72) /
100

# Find critical values
fp = sp.diff(f, x)
crits = []
for crit in sp.solve(fp, x):
  crits.append(crit.evalf())

print("Critical values at x = ", crits)

# Find increasing/decreasing intervals
if fp.subs(x, -4) > 0:
  print(f"f is increasing on (-inf, {crits[1]})")
else:
  print(f"f is decreasing on (-inf, {crits[1]})")
if fp.subs(x, -3) > 0:
  print(f"f is increasing on ({crits[1]}, {crits[2]})")
else:
  print(f"f is decreasing on ({crits[1]}, {crits[2]})")
if fp.subs(x, -2) > 0:
  print(f"f is increasing on ({crits[2]}, {crits[0]})")
else:
  print(f"f is decreasing on ({crits[2]}, {crits[0]})")
if fp.subs(x, 0.5) > 0:
  print(f"f is increasing on ({crits[0]}, {crits[3]})")
else:
  print(f"f is decreasing on ({crits[0]}, {crits[3]})")
if fp.subs(x, 0.66) > 0:
  print(f"f is increasing on ({crits[3]}, {crits[4]})")
else:
  print(f"f is decreasing on ({crits[3]}, {crits[4]})")
if fp.subs(x, 1) > 0:
  print(f"f is increasing on ({crits[4]}, {crits[5]})")
else:
  print(f"f is decreasing on ({crits[4]}, {crits[5]})")
```

```python
if fp.subs(x, 3) > 0:
  print(f"f is increasing on ({crits[5]}, inf)")
else:
  print(f"f is decreasing on ({crits[5]}, inf)")

# Plot f'
sp.plot(fp, xlim=[-4, 4], ylim=[-4, 4])
```

```
Critical values at x =  [0, -3.26323478808809, -2.55960783843188,
0.652860268750571, 0.679402180494846, 2.77629446298884]
f is decreasing on (-inf, -3.26323478808809)
f is increasing on (-3.26323478808809, -2.55960783843188)
f is decreasing on (-2.55960783843188, 0)
f is increasing on (0, 0.652860268750571)
f is decreasing on (0.652860268750571, 0.679402180494846)
f is increasing on (0.679402180494846, 2.77629446298884)
f is decreasing on (2.77629446298884, inf)
```

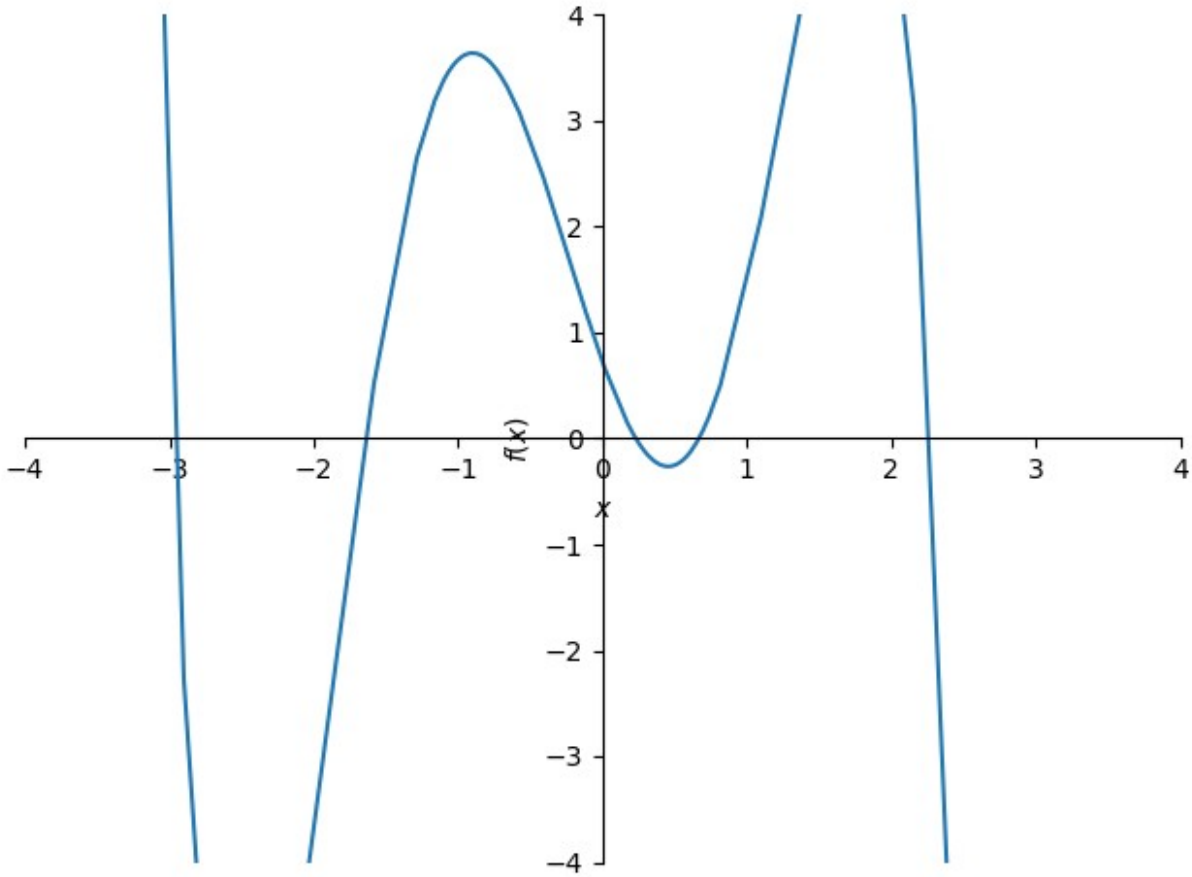



```
<sympy.plotting.plot.Plot at 0x10e5a0250>
```

1b

```python
# Find points of inflection
fpp = sp.diff(fp, x)
inflects = []
for inflect in sp.solve(fpp, x):
  inflects.append(inflect.evalf())

print("Inflection points at x = ", inflects)

# Find concave up/down intervals
if fpp.subs(x, -3) > 0:
  print(f"f is concave up on (-inf, {inflects[0]})")
else:
  print(f"f is concave down on (-inf, {inflects[0]})")
if fpp.subs(x, -2) > 0:
  print(f"f is concave up on ({inflects[0]}, {inflects[1]})")
else:
  print(f"f is concave down on ({inflects[0]}, {inflects[1]})")
if fpp.subs(x, 0) > 0:
  print(f"f is concave up on ({inflects[1]}, {inflects[2]})")
else:
  print(f"f is concave down on ({inflects[1]}, {inflects[2]})")
if fpp.subs(x, 0.5) > 0:
  print(f"f is concave up on ({inflects[2]}, {inflects[3]})")
else:
  print(f"f is concave down on ({inflects[2]}, {inflects[3]})")
if fpp.subs(x, 1) > 0:
  print(f"f is concave up on ({inflects[3]}, {inflects[4]})")
else:
  print(f"f is concave down on ({inflects[3]}, {inflects[4]})")
if fpp.subs(x, 3) > 0:
  print(f"f is concave up on ({inflects[4]}, inf)")
else:
  print(f"f is concave down on ({inflects[4]}, inf)")

# Plot f''
sp.plot(fpp, xlim=[-4, 4], ylim=[-4, 4])
```

```
Inflection points at x =  [-2.97484909632770, -1.63999934722655,
0.230329396036398, 0.666271349779803, 2.28967626916661]
f is concave up on (-inf, -2.97484909632770)
f is concave down on (-2.97484909632770, -1.63999934722655)
f is concave up on (-1.63999934722655, 0.230329396036398)
f is concave down on (0.230329396036398, 0.666271349779803)
f is concave up on (0.666271349779803, 2.28967626916661)
f is concave down on (2.28967626916661, inf)
```




```
<sympy.plotting.plot.Plot at 0x10e4520a0>
```

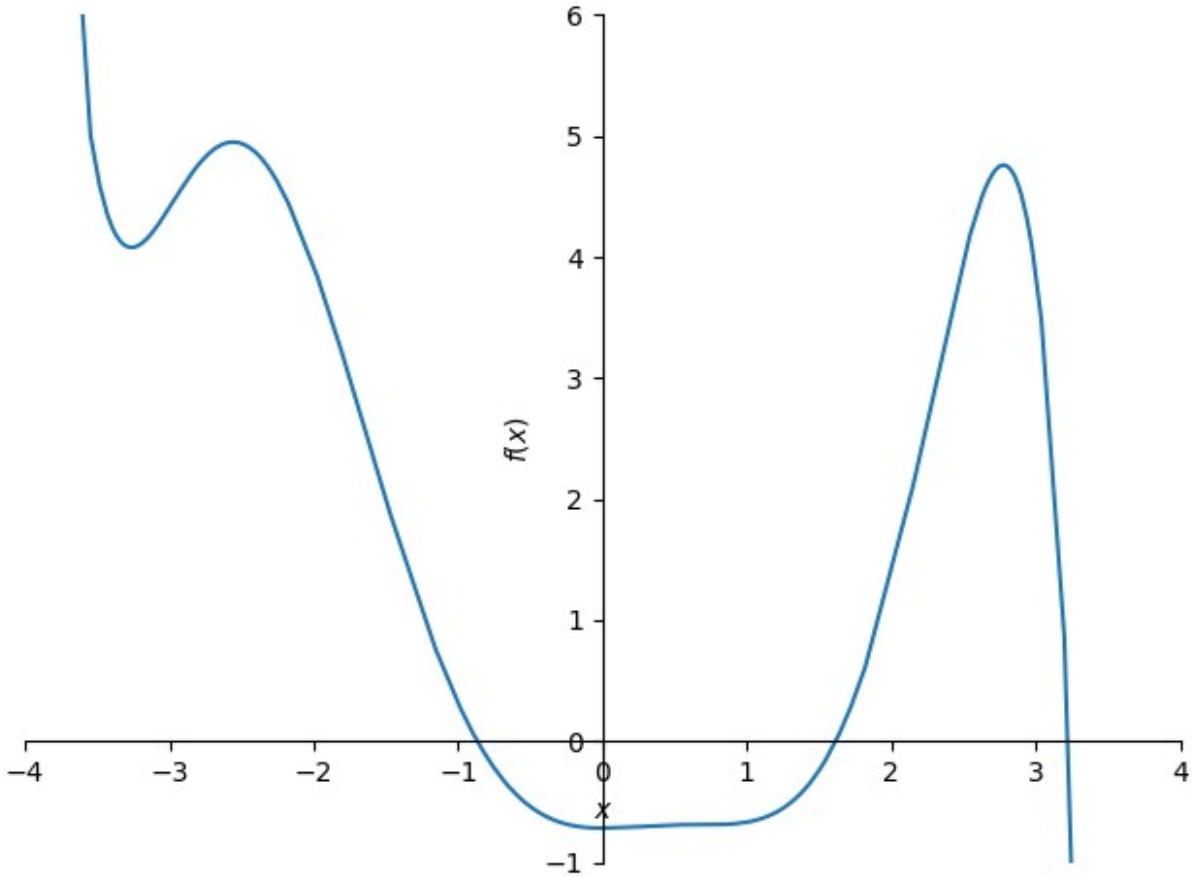
1c

```
# Enter your answer as a print statement or series of comments.
print(f"There are 6 local extrema points and 5 inflection points.")
```

```
There are 6 local extrema points and 5 inflection points.
```

1d

```
# Plot f
sp.plot(f, xlim=[-4, 4], ylim=[-1, 6])
```

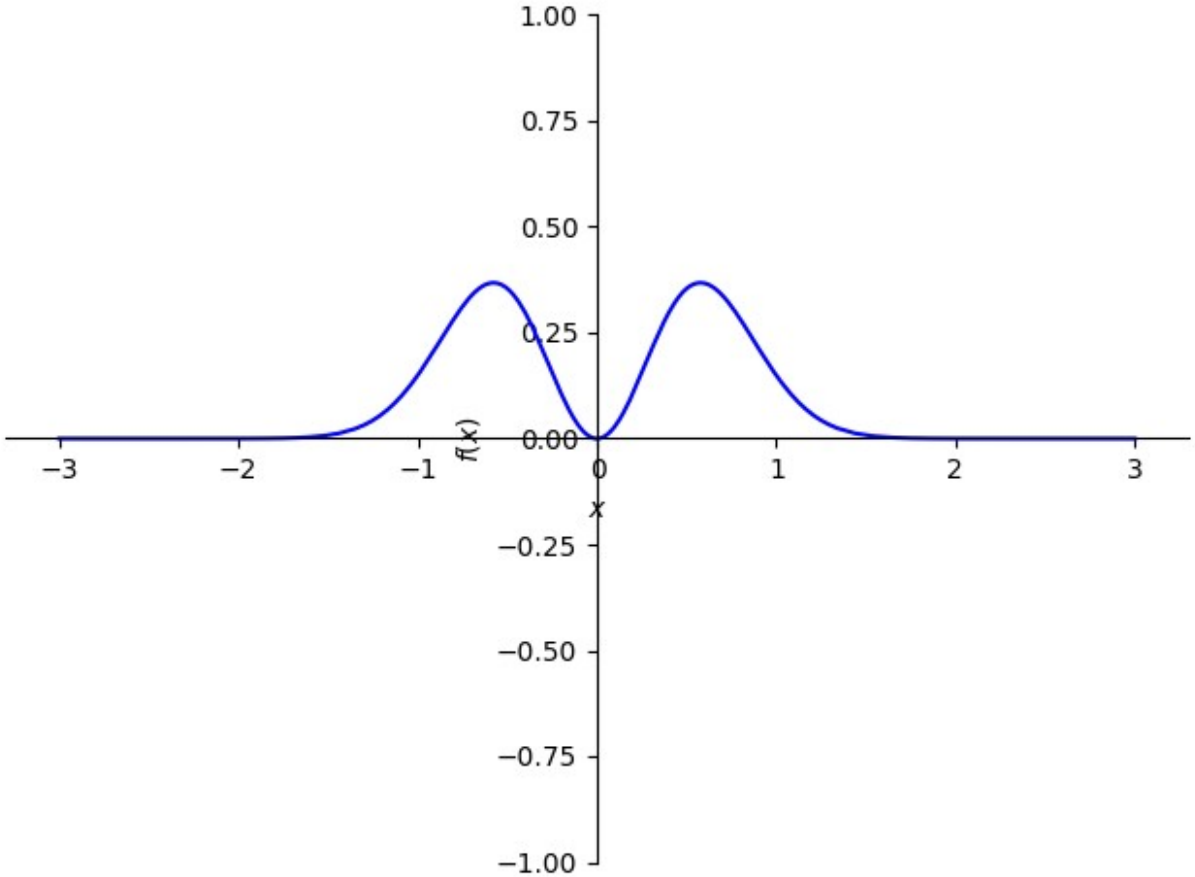



```
<sympy.plotting.plot.Plot at 0x10e45a160>
```

Question 2

2a

```
x, b = sp.symbols('x b')

g = b*x**2*sp.E**(-b*x**2)
for i in range(-3, 4):
  g_sub = g.subs({b: i})
  p = plot(g_sub, (x, -3, 3), ylim=[-1, 1], line_color = 'blue',
show=False)
p.show()
```



2b

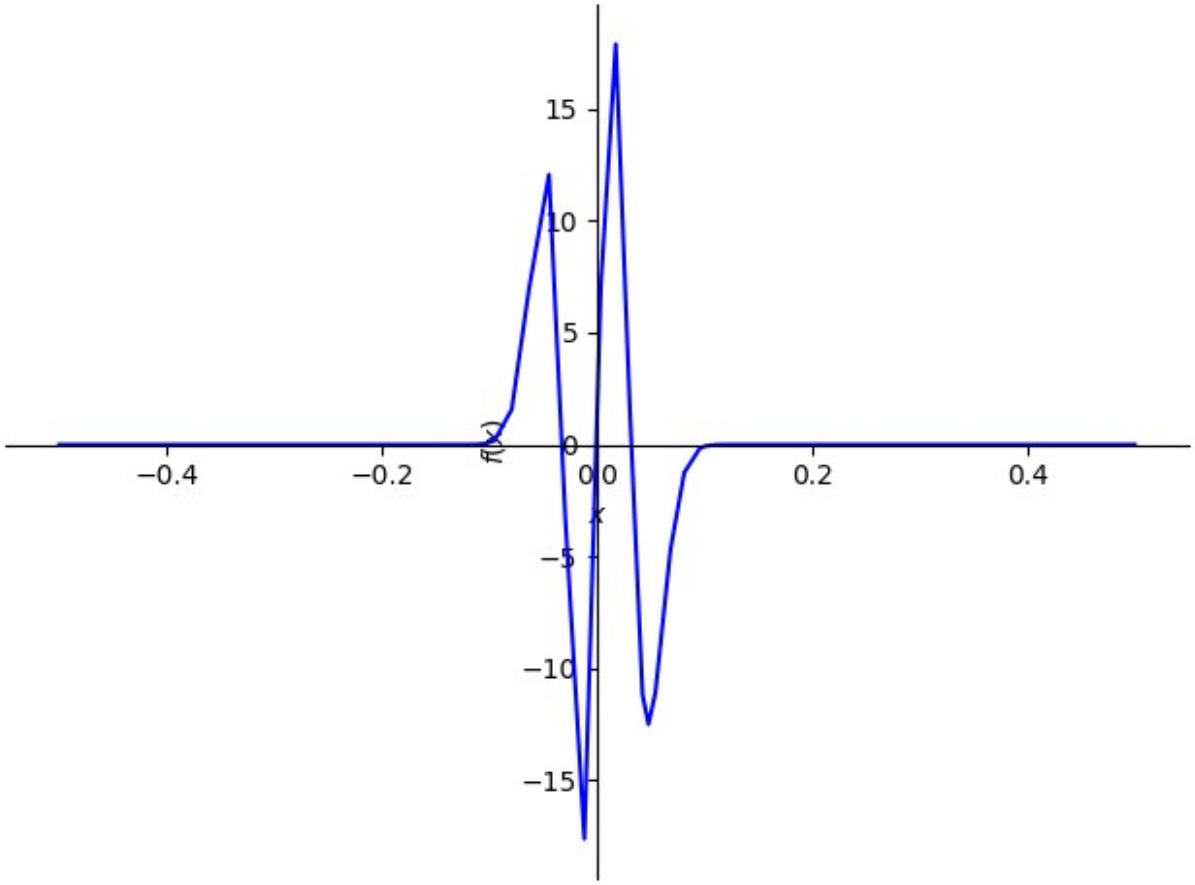
```
gprime = sp.diff(g, x)
cval = sp.solve(gprime, x)
print('There are critical values at x =', cval)
```

```
There are critical values at x = [0, -sqrt(1/b), sqrt(1/b)]
```

2c

```
print('The critical values as b goes to infinity all become 0')
gprime_sub = gprime.subs({b: 1000})
plot(gprime_sub, (x, -0.5, 0.5), line_color = 'blue')
```

```
The critical values as b goes to infinity all become 0
```




```
<sympy.plotting.plot.Plot at 0x11e3886a0>
```

2d

```
g2prime = sp.diff(gprime, x)
cval = sp.solve(g2prime, x)
print('There are points of inflection at x =', cval, 'when b != 0')
```

```
There are points of inflection at x = [-sqrt(5 -
sqrt(17))*sqrt(1/b)/2, sqrt(5 - sqrt(17))*sqrt(1/b)/2, -sqrt(sqrt(17)
+ 5)*sqrt(1/b)/2, sqrt(sqrt(17) + 5)*sqrt(1/b)/2] when b != 0
```

Question 3

3a

```
sigma, x = sp.symbols('σ x')

f = ( 1 / (sigma * (2 * sp.pi) ** 0.5) ) * sp.E ** ( sp.Rational(-1,
2) * (x / sigma)**2 )

for sigma_v in [1, 5, 10]:
    F = f.subs(sigma, sigma_v)
```

```
    dFdx = sp.diff(F, x)
    cvs = sp.solve(dFdx)
    print(f'When σ = {sigma_v}, the critical value is {cvs[0]:.2f}')
```

```
When σ = 1, the critical value is 0.00
When σ = 5, the critical value is 0.00
When σ = 10, the critical value is 0.00
```

3b

```
for sigma_v in [1, 5, 10]:
    F = f.subs(sigma, sigma_v)
    dFdx = sp.diff(F, x, 2)
    infpts_x = sp.solve(dFdx)
    infpts_y = [F.subs(x, x_v) for x_v in infpts_x]
    print(f'When σ = {sigma_v}, the inflection points are at
({infpts_x[0]:.2f}, {infpts_y[0]:.2f}) and ({infpts_x[1]:.2f},
{infpts_y[0]:.2f})')
```

```
When σ = 1, the inflection points are at (-1.00, 0.24) and (1.00,
0.24)
When σ = 5, the inflection points are at (-5.00, 0.05) and (5.00,
0.05)
When σ = 10, the inflection points are at (-10.00, 0.02) and (10.00,
0.02)
```

3c

The pattern is that, for a given σ, the x values that the inflection points occur at are σ and -σ.